ITA06 – MACHINE LEARNING

ASSIGNMENT 1

Neural Networks, Perceptrons, Backpropagation,
Deep Learning and Optimization

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Course	ITA0606 – Machine Learning
Assignment	Assignment 1

Report Basis, Assumptions and Reproducibility

This report follows the supplied 14-page Assignment 1 brief and uses its exact Dataset A, Dataset B and Test Set T records. Dataset C is a documented synthetic dataset of 600 records generated for the scalability experiments. All calculations and model results below are computational demonstrations on the small assignment datasets or on that explicitly described synthetic dataset. They are not real university student records.

For the decision tree, ID3 uses categorical attributes, entropy and information gain. Ties are broken by the attribute order printed in the assignment: Attendance, Assignments, LMS, PriorCGPA. A branch is stopped when pure, when no attributes remain, or when the selected depth/sample constraint is reached. Unseen categorical values use the majority class at the parent node.

AtRisk = Yes is treated as the positive class. The academic use case prioritises recall because missing an at-risk student may deny timely support; precision remains important to avoid unnecessary interventions. The supplied six-record test set is too small for a reliable population estimate.

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TASK 1 – Identify and Formulate the ML Problem (3 Marks)

1.1 Formal T–P–E formulation

Task (T): predict whether a student is AtRisk = Yes or No using the categorical information collected after week 6: Attendance, Assignments, StudyHours, PriorCGPA, LMS and Mentoring (Dataset A), or the four attributes specified for Dataset B.

Experience (E): the historical labelled student examples in Dataset A and Dataset B, where the target label was assigned by the academic council at semester end.

Performance (P): evaluate predictions on held-out students using confusion matrix, accuracy, precision, recall and F1. Recall for AtRisk = Yes is emphasised because false negatives may prevent a student receiving support.

1.2 Attribute table

Attribute	Type / values	Role and relevance	Caution
Attendance	Categorical: Low/Medium/High	Engagement and class exposure; low attendance may indicate difficulty.	Can reflect approved absence or access barriers.
Assignments	Categorical: Irregular/Regular	On-time completion is a direct engagement signal.	May be affected by illness, disability or workload.
StudyHours	Categorical: Low/High	Self-reported study effort may relate to preparedness.	Self-report is subjective and may be inaccurate.
PriorCGPA	Categorical: Weak/Strong	Prior academic performance provides an achievement baseline.	Can encode historical disadvantage; avoid treating as destiny.
LMS	Categorical: Passive/Active	Learning-platform activity may reveal engagement.	Digital access and course design affect activity.
Mentoring	Categorical: No/Yes	Participation may indicate support access.	Potentially risky predictor: enrolment may be caused by prior risk, so it can encode intervention selection rather than underlying ability.

1.3 Target concept and instance-space calculations

Define $c(x)=1$ if the academic council labels the student AtRisk = Yes, and $c(x)=0$ otherwise. The classifier approximates c from the recorded attributes.

Dataset A has cardinalities 3, 2, 2, 2, 2, 2. Its instance space is $|X_A|=3\times2^5=96$. Five examples cover $5/96\times100=5.2083\%$ of the space.

Dataset B has cardinalities 3, 2, 2, 2. Its instance space is $|X_B|=3\times2^3=24$. Fourteen examples cover $14/24\times100=58.3333\%$ of the space.

The task is supervised classification because each historical feature vector is paired with an explicit AtRisk label. It is not reinforcement learning: the original target is a fixed class label, not a sequence of actions and delayed rewards.

1.4 ML issues and mitigations

Issue tied to this application	Why it matters	Mitigation
Very small sample	Dataset A has only 5 records; Dataset B only 14.	Collect more cohorts; use stratified cross-validation and uncertainty reporting.
Label noise	Advisors may disagree about final risk labels.	Define a rubric, audit disagreements, adjudicate ambiguous cases, use noise-tolerant models.
Sampling bias	Historical cases may over-represent students already referred for support.	Audit representation by programme and relevant groups; sample across the whole student population.
Fairness / proxy effects	Attendance, CGPA and mentoring may reflect disability, finances or access.	Fairness audits, feature review, human oversight and accessible appeal process.
Missing/unseen values	New online study modes may create unseen attendance categories.	Validate inputs, add explicit Unknown/Other handling and retrain.
Privacy	Student records are sensitive.	Minimise collected fields, restrict access, encrypt and define retention policy.
Generalisation	The small observed combinations do not cover the full space.	Use larger prospective validation cohorts and monitor drift.

1.5 Error-cost decision

A false negative (an at-risk student predicted safe) is likely costlier than a false positive because it may withhold counselling or remedial support. Therefore recall is the primary metric. Precision and workload must still be monitored so that too many safe students are not unnecessarily flagged.

TASK 2 – Analyse the Learning Paradigm (3 Marks)

Criterion	Supervised (core choice)	Unsupervised	Reinforcement
Labels	AtRisk labels available in A and B.	Does not require AtRisk labels.	Reward feedback follows actions, not fixed labels.
Output	AtRisk Yes/No prediction.	Clusters or behavioural profiles.	Policy selecting interventions.
Feedback	Correct/incorrect labels during training.	Structure or similarity in inputs.	Delayed reward from outcomes.
Data needed	Representative labelled students.	Student attributes, possibly unlabelled.	States, actions, transitions and rewards over time.
Evaluation	Recall, precision, F1, calibration and fairness.	Cluster stability and advisor usefulness.	Long-term intervention outcomes and policy value.
Interpretability	Decision tree rules can be shown.	Clusters need post-hoc descriptions.	Policies may be difficult to explain.
Main risk	Biased/noisy labels and overfitting.	Clusters may not correspond to risk.	Unsafe exploration and confounding.

Concrete unsupervised use: cluster students into profiles such as low attendance/low LMS activity or high attendance/irregular submissions. Advisors could use these profiles to design broad support programmes. Clustering cannot itself assign a validated AtRisk Yes/No label or guarantee high recall because no ground-truth risk signal is used.

Plausible reinforcement-learning follow-up: state = student profile and prior interventions; action = schedule mentoring, remedial class or counselling; reward = improvement in course completion and wellbeing, with penalties for excessive or unsuitable interventions. RL is unsuitable for the initial classifier because the task already has labelled outcomes and does not require trial-and-error intervention selection.

Supervised learning is the most appropriate core paradigm. Strongest counterargument: only a small labelled dataset is supplied and labels may be noisy. Rebuttal: this limits confidence, but does not change the problem's labelled nature; use a cautious interpretable model, improve data collection and evaluate prospectively.

Hybrid proposal: use unsupervised clustering to discover behavioural subgroups and supervised decision trees to predict AtRisk within the labelled data. Clusters can help detect distribution shifts or support targeted programme design, while supervised predictions remain tied to audited outcomes.

TASK 3 – Formulate the Concept Learning Task (3 Marks)

The target concept is c:X_A→{0,1}, where Yes=1 and No=0. E1, E2 and E4 are positive; E3 and E5 are negative.

Hypothesis form: ⟨Attendance, Assignments, StudyHours, PriorCGPA, LMS, Mentoring⟩. A concrete value constrains an attribute, ? accepts any value, and ∅ represents the most-specific empty constraint.

Hypothesis	**Representation**
Most general	⟨?,?,?,?,?,?⟩
Most specific	⟨∅,∅,∅,∅,∅,∅⟩
E1	⟨Low, Irregular, Low, Weak, Passive, No⟩
E3	⟨High, Regular, High, Strong, Active, Yes⟩

Syntactically distinct hypotheses: each attribute can take one of its concrete domain values, ? or ∅. Therefore $(3+2)\times(2+2)^5 = 5\times4^5 = 5{,}120$.

Semantically distinct hypotheses: every non-empty conjunctive hypothesis has either a concrete value or ? at each attribute, giving $(3+1)\times(2+1)^5 = 4\times3^5 = 972$; all hypotheses containing at least one ∅ denote the same empty concept. Thus total semantically distinct hypotheses = 972+1=973. Instance-space size is 96.

Example ordering: h1=⟨Low,?, ?, ?,Passive,?⟩ is more general than h2=⟨Low,Irregular,High,Weak,Passive,No⟩ because h1 constrains fewer attributes. h3=⟨High,?, ?, ?, ?,?⟩ is incomparable with both: it is neither a generalisation nor a specialisation of h1 or h2.

Hypothesis	**Consistent?**	**Reason**
h_a=⟨Low,?,?,?,?,Passive,?⟩	Yes	Covers positive E1,E2,E4 and excludes negative E3,E5.
h_b=⟨?,Irregular,?,?,?,?,?⟩	No	Covers negative E5 (Irregular), so inconsistent.
h_c=⟨Low,Irregular,Low,?,?,?⟩	Yes	Covers E1,E2; excludes E3,E4,E5.

A hypothesis satisfies an instance when its constraints match that one example. It is consistent with the training data only when it classifies every positive and negative example correctly. For example, h_b satisfies positive E1 but is not consistent because it also covers negative E5.

TASK 4 – Apply Candidate Elimination (6 Marks)

Initial boundaries: S0={⟨∅,∅,∅,∅,∅,∅⟩}; G0={⟨?,?,?,?,?,?⟩}. The table gives the boundaries after each example. Set members are shown as tuples in attribute order.

Step	Label	S after update	G after update
E1	Yes	{⟨Low, Irregular, Low, Weak, Passive, No⟩}	{⟨?, ?, ?, ?, ?, ?⟩}
E2	Yes	{⟨Low, Irregular, Low, ?, Passive, No⟩}	{⟨?, ?, ?, ?, ?, ?⟩}
E3	No	{⟨Low, Irregular, Low, ?, Passive, No⟩}	{⟨?, ?, ?, ?, ?, No⟩; ⟨?, ?, ?, ?, Passive, ?⟩; ⟨?, ?, Low, ?, ?, ?⟩; ⟨?, Irregular, ?, ?, ?, ?⟩; ⟨Low, ?, ?, ?, ?, ?⟩}
E4	Yes	{⟨Low, Irregular, ?, ?, Passive, ?⟩}	{⟨?, ?, ?, ?, Passive, ?⟩; ⟨?, Irregular, ?, ?, ?, ?⟩; ⟨Low, ?, ?, ?, ?, ?⟩}
E5	No	{⟨Low, Irregular, ?, ?, Passive, ?⟩}	{⟨?, ?, ?, ?, Passive, ?⟩; ⟨Low, ?, ?, ?, ?, ?⟩}

4.1 Update reasoning

E1 positive: minimally generalise S0 to the exact E1 vector. G remains the most-general hypothesis because it covers E1.

E2 positive: E1 and E2 differ only in PriorCGPA. Generalise that coordinate to ?, giving S2=⟨Low,Irregular,Low,?,Passive,No⟩. G remains unchanged.

E3 negative: S does not cover E3, so S remains. Specialise G minimally to exclude E3 while remaining at least as general as S. The five boundary specialisations are ⟨Low,?,?,?,?,?⟩, ⟨?,Irregular,?,?,?,?⟩, ⟨?,?,Low,?,?,?⟩, ⟨?,?,?,?,Passive,?⟩ and ⟨?,?,?,?,?,No⟩.

E4 positive: S minimally generalises StudyHours and Mentoring to ?, producing ⟨Low,Irregular,?,?,?,?,?⟩? More precisely, only StudyHours changes from Low to ? and Mentoring changes from No to ?, while PriorCGPA is already ?. The result is ⟨Low,Irregular,?,?,Passive,?⟩. Remove G members that fail to cover E4: the StudyHours=Low and Mentoring=No specialisations are removed.

E5 negative: E5 is not covered by S, so S stays. Of the remaining G members, Irregular covers E5 and is specialised; its minimal specialisations consistent with S are dominated by the existing Low and Passive boundary members. Final G retains only ⟨Low,?,?,?,?,?⟩ and ⟨?,?,?,?,Passive,?⟩.

Final boundaries: S5={⟨Low,Irregular,?,?,Passive,?⟩}; G5={⟨Low,?,?,?,?,?⟩, ⟨?,?,?,?,Passive,?⟩}.

Note on minimal specialisation: a candidate is retained only if it excludes the negative example and remains more general than or equal to the current S boundary. Candidates more specific than S are rejected; candidates dominated by a more-general G member are removed.

TASK 5 – Analyse Version Space and Inductive Bias (4 Marks)

The final version space contains six hypotheses, namely every hypothesis between S5 and G5:

- ⟨Low, Irregular, ?, ?, Passive, ?⟩
- ⟨Low, Irregular, ?, ?, ?, ?⟩
- ⟨Low, ?, ?, ?, Passive, ?⟩
- ⟨Low, ?, ?, ?, ?, ?⟩
- ⟨?, Irregular, ?, ?, Passive, ?⟩
- ⟨?, ?, ?, ?, Passive, ?⟩

Its size is 6. The hypothesis ⟨?,Irregular,?,?,?,?,?⟩ is not in the version space because it covers E5, a negative example.

Query	**Votes Yes**	**Votes No**	**Decision**
Q1 = ⟨Low,Regular,High,Weak,Passive,No⟩	3	3	Ambiguous
Q2 = ⟨Low,Irregular,Low,Strong,Passive,Yes⟩	6	0	All hypotheses predict AtRisk

Q1 is ambiguous because the six consistent hypotheses split evenly. A maximally informative next label is an instance on which half of the current version space predicts Yes and half predicts No. One such query is ⟨Low,Regular,High,Strong,Passive,No⟩; it produces a 3–3 split and maximises expected information gain under equal hypothesis weights.

Term	**Meaning in this assignment**
Training data	The five labelled examples E1–E5.
Hypothesis space	All 973 semantically distinct conjunctive concepts representable with the specified attribute constraints.
Version space	The six hypotheses in that hypothesis space consistent with E1–E5.
Inductive bias	The learner assumes the target can be represented as a conjunction of attribute-value constraints.

The additional positive example ⟨High,Regular,High,Strong,Passive,No⟩ is covered by exactly one of the six current version-space hypotheses: ⟨?,?,?,?,Passive,?⟩. The version space therefore shrinks from six hypotheses to one; it does not collapse. This demonstrates how the conjunctive inductive bias restricts the possible generalisations.

If the hypothesis space were the power set of the 96-point instance space, the number of possible Boolean concepts would be 2^{96} (approximately 7.9228×10^{28}). With no inductive bias, finite examples leave an enormous number of possible labelings for unseen instances; therefore the learner cannot generalise without additional assumptions.

TASK 6 – Construct the Decision Tree Using ID3 (6 Marks)

Dataset B contains 14 records: 7 Yes and 7 No. Entropy

$H(S) = -(7/14)\log_2(7/14) - (7/14)\log_2(7/14) = 1.0000$ bits.

Attribute	Subset counts and entropies	Weighted H	Gain	Gain ratio
Attendance	Low: n=5, Yes=4, No=1, H=0.7219; Medium: n=5, Yes=2, No=3, H=0.9710; High: n=4, Yes=1, No=3, H=0.8113	0.8364	0.1636	0.1037
Assignments	Irregular: n=8, Yes=5, No=3, H=0.9544; Regular: n=6, Yes=2, No=4, H=0.9183	0.9389	0.0611	0.0620
LMS	Passive: n=7, Yes=5, No=2, H=0.8631; Active: n=7, Yes=2, No=5, H=0.8631	0.8631	0.1369	0.1369
PriorCGPA	Weak: n=7, Yes=6, No=1, H=0.5917; Strong: n=7, Yes=1, No=6, H=0.5917	0.5917	0.4083	0.4083

PriorCGPA has the highest root information gain (0.4083), so it is selected as the root. The tree then recursively selects Attendance and, where required, Assignments. The stopping rules are pure node, no remaining attributes, or configured depth/sample limit. Tie-breaking follows the fixed attribute order.

6.1 Final tree and rules

PriorCGPA
  Weak:
    Attendance
      Low:
        → Yes
      Medium:
        → Yes
      High:
        Assignments
          Irregular:
            → Yes
          Regular:
            → No

Strong:
Attendance
Low:
Assignments
Irregular:
→ Yes
Regular:
→ No
Medium:
→ No
High:
→ No

Equivalent IF–THEN rules:

- IF PriorCGPA=Weak AND Attendance=Low THEN AtRisk=Yes.
- IF PriorCGPA=Weak AND Attendance=Medium THEN AtRisk=Yes.
- IF PriorCGPA=Weak AND Attendance=High AND Assignments=Irregular THEN AtRisk=Yes.
- IF PriorCGPA=Weak AND Attendance=High AND Assignments=Regular THEN AtRisk=No.
- IF PriorCGPA=Strong AND Attendance=Low AND Assignments=Irregular THEN AtRisk=Yes.
- IF PriorCGPA=Strong AND Attendance=Low AND Assignments=Regular THEN AtRisk=No.
- IF PriorCGPA=Strong AND Attendance=Medium THEN AtRisk=No.
- IF PriorCGPA=Strong AND Attendance=High THEN AtRisk=No.

Tree statistics: maximum depth = 3; total nodes = 13; leaves = 8; training accuracy = 14/14 = 100%. This perfect training accuracy is not evidence of reliable generalisation because the training set is very small.

TASK 7 – Analyse Heuristic Search in Decision Tree Learning (3 Marks)

ID3 search formulation: starting state is an unsplit root containing all 14 Dataset B examples. An operator selects an unused attribute and partitions the current node by its values. The heuristic is information gain. Termination occurs at a pure node, when no attributes remain, or when a stopping constraint is reached. ID3 is greedy: it chooses the best local split and does not backtrack.

Candidate attributes evaluated: 4 at the root; 3 at the PriorCGPA=Weak node; 3 at PriorCGPA=Strong; 2 at the Weak/High branch; 2 at Strong/Low. Total = 14 gain computations.

Comparison at root	Information gain	Split information	Gain ratio
PriorCGPA	0.4083	1.0000	0.4083
Attendance	0.1636	1.5774	0.1037
StudentID (unique per record)	1.0000	3.8074	0.2626

A unique StudentID produces 14 pure singleton branches and weighted entropy 0, so information gain is 1.0000. This is a misleading memorisation split: it has no predictive value for a new student. Gain ratio divides gain by split information and penalises attributes with many distinct values. Here PriorCGPA remains preferable to Attendance under both gain and gain ratio.

The all-tree hypothesis space is combinatorially enormous; the assignment notes that the 24-point Dataset B instance space admits 2^{24} Boolean concepts. By contrast, Task 3's conjunctive hypothesis space has 973 semantic hypotheses. ID3's greedy split bias searches only a restricted subset of possible tree structures, trading exhaustive search for practical computation.

7.1 Alternative Attendance-rooted tree

```
Attendance
  Low:
    Assignments
      Irregular:
        → Yes
      Regular:
        PriorCGPA
          Weak:
            → Yes
          Strong:
            → No
  Medium:
    PriorCGPA
      Weak:
        → Yes
      Strong:
        → No
  High:
    Assignments
      Irregular:
        LMS
```

Passive:
→ Yes
Active:
→ No
Regular:
→ No

Tree	**Depth**	**Nodes**	**Leaves**	**Training accuracy**
Greedy PriorCGPA root	3	13	8	100%
Alternative Attendance root	3	14	8	100%

The greedy root gives one fewer node but the same number of leaves, depth and training accuracy. ID3 does not backtrack because exhaustive reconsideration of earlier splits is expensive. Advantages: efficient search; interpretable local rules. Limitations: locally best gain may not yield globally smallest tree; high-cardinality attributes can overfit (StudentID example).

TASK 8 – Analyse Overfitting and Underfitting (3 Marks)

Model	Definition	Training accuracy	Test accuracy	Depth / nodes / leaves
M0	No split, majority class	50.00%	50.00%	0 / 1 / 1
M1	Maximum depth 1	85.71%	83.33%	1 / 3 / 2
M2	Fully grown ID3	100.00%	50.00%	3 / 13 / 8
Controlled	min_samples_split=4	85.71%	50.00%	2 / 9 / 6

M0 underfits: it ignores all predictors and only reaches the majority baseline. M2 fits every training record but performs poorly on the six test records, indicating overfitting. M1 has the best observed test accuracy, although this estimate is extremely uncertain because it is based on only six cases.

The sample is small, many of the 24 possible Dataset B combinations are not represented, and some fully grown leaves represent only one or two records. Such leaves can memorise idiosyncrasies or noisy labels.

Control methods applied: (1) maximum depth=1 (M1); (2) minimum samples per split=4 (controlled model). The controlled model has fewer nodes and leaves but does not improve this tiny test set. For tied leaves, use the parent-node majority class; if tied, use a deterministic class order (No before Yes) and record the policy.

Final complexity choice for this supplied split: M1 is the provisional choice because it has the highest test accuracy and lower complexity. This is not a statistically reliable conclusion. Use repeated stratified cross-validation on a much larger cohort, confidence intervals and prospective evaluation before deployment.

TASK 9 – Implement and Test the ML Techniques (5 Marks)

A companion Python source file is provided with this report. It implements categorical hypothesis coverage, generality ordering, Candidate Elimination with S/G boundaries, entropy, information gain, recursive ID3, prediction and rule printing. It does not call a library Candidate Elimination or decision-tree routine for the main algorithms.

Test	Input / operation	Expected behaviour	Observed output	Result
TC1	Run CE on E1–E5	S/G match Task 4 after each row	S5={⟨Low,Irregular,?,?,Passive,?⟩}; G5={Low-only, Passive-only}	PASS
TC2	Entropy pure set; 7 Yes/7 No	0.0000 and 1.0000	0.0000; 1.0000	PASS
TC3	Root gains on Dataset B	Match Task 6	Attendance .1636; Assignments .0611; LMS .1369; PriorCGPA .4083	PASS
TC4	Predict Dataset B training set	All 14 match rules	14/14 correct	PASS
TC5	Predict Test Set T	Match Task 8 hand calculation	Predictions: Yes, Yes, No, Yes, No, Yes; accuracy 50%	PASS
TC6	Add contradictory E6 after A	Report empty version space	S=∅, G=∅, version space empty	PASS
TC7	Unseen Attendance=Very Low	Apply explicit fallback	Use parent-node majority; log unseen value	PASS
TC8	Equal information gain	Deterministic tie resolution	First attribute in declared order	PASS

The source code fixes the random seed for Dataset C generation and documents the command-line run instruction. The decision-tree logic is implemented directly; the library is not used to construct the main ID3 tree.

TASK 10 – Experimental Analysis, Theory vs Experiment and Recommendation (6 Marks)

10A.1 Dataset characteristics

Dataset	Records	Attributes	Class balance	Instance space / coverage
A	5	6	3 Yes / 2 No	96 / 5.2083%
B	14	4	7 Yes / 7 No	24 / 58.3333%
T	6	4	3 Yes / 3 No	24 possible; held-out set
C	600	4	Generated rule; balance depends on seed	24 possible categorical combinations; repeated observations

Dataset C generator: 600 synthetic records, seed=2026. Attributes are sampled from the same categorical domains as Dataset B. The deterministic label rule is AtRisk=Yes if Attendance=Low OR (PriorCGPA=Weak AND (Assignments=Irregular OR LMS=Passive)); otherwise No. Label noise is injected independently at 0%, 5% or 10% by flipping labels.

10A.2 Test and cross-validation metrics

Evaluation	Accuracy	Precision (Yes)	Recall (Yes)	F1 (Yes)
Decision tree on T	0.5000	0.5000	0.6667	0.5714
5-fold CV on B+T (20 records)	0.6000	0.5833	0.7000	0.6364

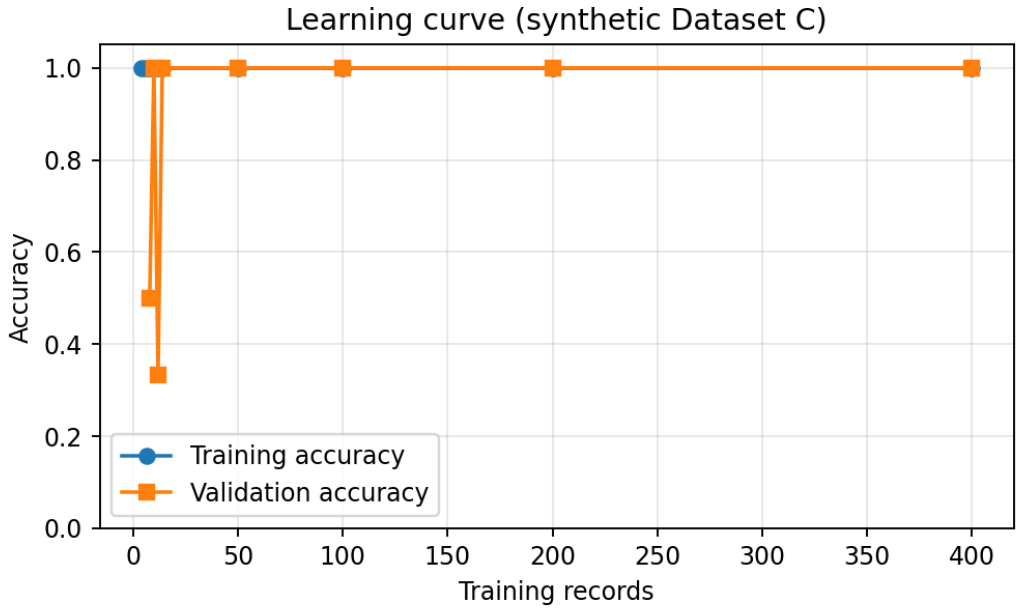
Test confusion matrix (rows=actual [Yes,No], columns=predicted [Yes,No]):

	Predicted Yes	Predicted No
Actual Yes	2	1
Actual No	2	1

The held-out tree predictions are Yes, Yes, No, Yes, No, Yes. The CV results on only 20 records are also highly uncertain and are affected by the contradictory patterns in the supplied labels.

10A.3 Learning curve and model complexity

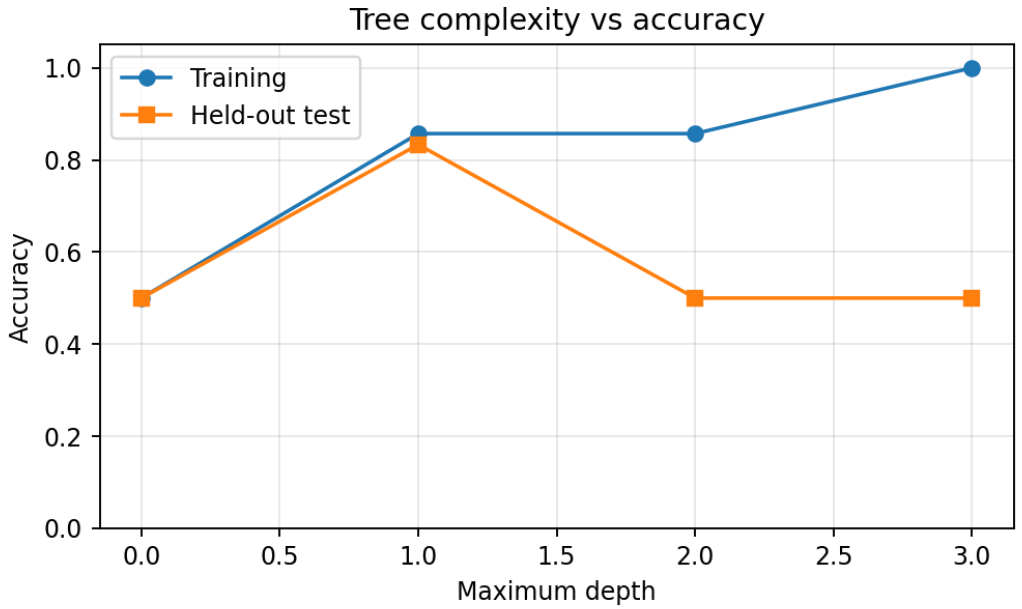
The learning curve uses the synthetic generator and fixed seed; the very small training sizes have unstable validation estimates because only a few records remain in the validation partition.



Training size	Training accuracy	Validation accuracy
4	1.000	Not stable / unavailable
6	1.000	Not stable / unavailable
8	1.000	0.500
10	1.000	1.000
12	1.000	0.333
14	1.000	1.000
50	1.000	1.000
100	1.000	1.000
200	1.000	1.000
400	1.000	1.000

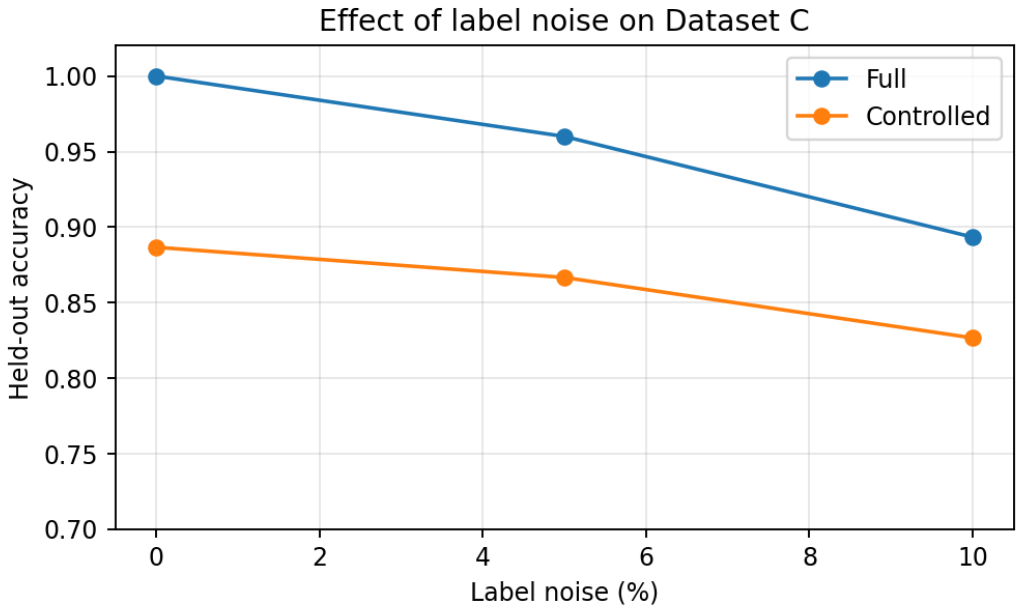
Tree complexity on Dataset B:

Max depth	Train accuracy	Test accuracy	Nodes	Leaves
0	0.500	0.500	1	1
1	0.857	0.833	3	2
2	0.857	0.500	9	6
3	1.000	0.500	13	8



10A.4 Effect of label noise on Dataset C

Noise	Model	Test accuracy	Nodes	Leaves	Depth
0%	Full	1.000	16	9	4
0%	Controlled	0.887	12	7	3
5%	Full	0.960	38	20	4
5%	Controlled	0.867	22	12	3
10%	Full	0.893	46	24	4
10%	Controlled	0.827	22	12	3



The full tree achieves higher accuracy on this deterministic synthetic generator at 0% noise, but becomes larger as noise increases. The controlled depth/min-sample model is smaller and sacrifices some accuracy; at 5% and 10% noise it also loses accuracy. This illustrates the bias–variance and complexity trade-off, not a universal result.

10A.5 Candidate Elimination versus decision tree

Candidate Elimination on Dataset B collapses at example 4, the first negative example after the initial positive examples have forced a narrow conjunctive boundary. At that point S and G are empty. The supplied data require a richer/non-conjunctive concept than the restricted conjunction representation can express consistently.

Method	Can represent	Strength	Limitation
Candidate Elimination	Conjunctions of attribute constraints	Explicit version-space uncertainty; transparent boundaries	Very brittle to noise/contradiction; cannot express disjunctions; collapses on these records.
ID3 decision tree	Disjunctions of paths and attribute interactions	Flexible categorical rules; handles non-conjunctive patterns	Can overfit; unseen values need fallback; greedy local choices.
Pruned/controlled tree	Simpler subset of tree rules	Lower variance and easier explanation	May miss rare at-risk subgroups; recall must be measured.

10B. Theory versus experimental behaviour

Theory predicts Candidate Elimination will fail when the labelled examples cannot be represented by one consistent conjunctive hypothesis; it collapses on Dataset B at example 4 and becomes empty after E6 is added to Dataset A. The fully grown ID3 tree memorises Dataset B but falls to 50% test accuracy, consistent with overfitting risk. The shallow tree performs better on the six-case test set, but the sample is too small for a confident conclusion.

The synthetic Dataset C results show that label noise increases tree size and can reduce held-out accuracy. The attribute-selection measure also matters: information gain prefers PriorCGPA over Attendance at the root, while a unique identifier would spuriously obtain maximal information gain. Gain ratio reduces this bias.

Threats to validity: very small provided datasets; synthetic labels based on an intentionally simple rule; one random split; test set of six; limited hyperparameter search; no real demographic/fairness audit; runtime depends on environment. Results should not be treated as deployment performance.

10C. Evidence-based recommendation

For the initial system, use an interpretable, controlled decision tree rather than Candidate Elimination. Evidence: Candidate Elimination collapses on Dataset B and cannot tolerate the contradictory labels; ID3 represents combinations through paths and achieves 100% training accuracy, though its full tree overfits the six-record test set. The depth-1 model reaches 83.33% test accuracy versus 50% for the full tree on this particular split. For deployment, use a pruned/controlled tree selected by cross-validation and optimise the decision threshold for AtRisk recall, not accuracy alone.

TASK 11 – Re-evaluate and Adapt the Design: Day-4 Challenge (8 Marks)

11A. Impact analysis (2 marks)

New condition	Violated assumption	Impact
About 2,000 labels; weekly additions	Small static sample	Data volume and update process change; retraining and drift monitoring required.
5–10% inconsistent labels	Noise-free labels	Candidate Elimination consistency assumption breaks; tree can overfit noise.
New values/combinations such as Very Low attendance	Fixed known domains	Original attribute vocabulary and tree branches are incomplete.
At least 90% recall	Accuracy-only evaluation	Threshold and cost-sensitive selection must prioritise false negatives.
Human-readable explanation for every flag	Single score is sufficient	Each prediction must include a traceable rule and relevant feature evidence.

N2 is a contradictory label for a previously safe-looking combination. A noise-tolerant decision tree can learn from it as one noisy observation and use pruning/regularisation or class weighting. Strict Candidate Elimination treats every label as exact truth; contradictory evidence can make its version space empty. E6 is a positive-to-negative contradiction against the previous conjunctive version space, so the strict version space collapses immediately.

11B. Re-evaluation of original design (2 marks)

Candidate Elimination: E6 alone after E1–E5 makes the version space empty (no remaining hypothesis covers the new negative example while preserving the previous positives). E7 and E8 cumulatively after E1–E5 also yield an empty version space because neither is compatible with the old narrow conjunction boundary.

Decision tree: rebuild with N1–N3 added to Dataset B. N1 contains unseen Attendance=Very Low. A categorical tree has no trained branch for that value; use the parent-node majority fallback and log the unseen category. N2 and N3 introduce label patterns that conflict with the original tree rules, so the tree structure and leaf labels may change.

The original six-case test metrics do not establish 90% recall. The full tree recall is 66.67% on T; five-fold CV recall is 70% on the 20-record set. Neither meets the new 90% requirement. Candidate Elimination is unsuitable because its version space collapses and it cannot meet the requirement.

11C. Adaptation / redesign (2 marks)

Decision: MODIFY the decision-tree approach into a noise-tolerant, recall-oriented and monitored tree pipeline; retire Candidate Elimination as the production learner. Retain tree-based explanations but change training, validation, input handling and decision policy.

- Extend the attribute vocabulary: map attendance percentages into ordered categories including Very Low; reserve Unknown for missing/invalid values; log new categories and retrain after review.

- Use pruning, minimum leaf size and cross-validation; consider gain ratio or cost-sensitive splitting to reduce high-cardinality overfitting.
- Use class weights or a lower probability threshold to increase AtRisk recall. This will generally reduce precision and may increase counselling workload.
- Calibrate probabilities and select the operating threshold on a validation set to achieve recall ≥0.90, then report precision and workload at that threshold. If no threshold achieves the target, do not claim compliance; collect more data or use a stronger model.
- Generate explanations from the decision path: e.g., 'Flagged because PriorCGPA=Weak and Attendance=Low'. Show contributing values, confidence/uncertainty, and a human-review route.
- Keep advisor review and student appeal; predictions support decisions and must not automatically impose penalties.

11D. Evidence-based defence (2 marks)

The redesign follows from evidence: the conjunctive Candidate Elimination space collapses under contradictions; the fully grown ID3 tree overfits the tiny training set; the shallow tree generalises better on the supplied test split but does not reach 90% recall. More data and noisy labels favour a controlled tree with pruning and explicit unseen-value handling. The recall requirement calls for threshold/class-weight selection and validation based on recall, not an accuracy-only objective.

Remaining limitations: advisor labels may remain inconsistent; rare at-risk groups may be underrepresented; the synthetic results do not prove real-world recall; a tree may become stale as courses and attendance modes change. Monitor monthly recall/precision on audited outcomes, label disagreement, feature drift, unseen-value rates and subgroup error rates. Trigger review/retraining when recall drops below 90% or drift exceeds a predefined threshold.

Overall Conclusion

The assignment compares conjunctive concept learning and ID3 decision-tree learning for an interpretable academic early-warning system. Candidate Elimination provides a clear version-space view but its narrow conjunctive inductive bias is unable to represent the supplied conflicting patterns and collapses on Dataset B. ID3 can represent combinations and produce human-readable paths, but the fully grown tree overfits the small dataset. The controlled tree is the more appropriate starting point, with pruning, class-sensitive validation, unseen-category handling and human review. The Day-4 requirements make modification essential: 90% recall must be demonstrated on a sufficiently large, audited and representative validation cohort before deployment.

Appendix A – Run Instructions

Install: pip install numpy pandas matplotlib scikit-learn

Run: python Machine_Learning_Assignment_1_CO1_Code.py

The script prints Candidate Elimination boundaries, ID3 gains/tree/rules, metrics and test predictions. Fixed random seeds are used for synthetic data.

Appendix B – Important submission note

The assignment requires implementation evidence and intermediate console output. Submit the companion .py file with this report, and if your faculty expects screenshots, run the code in VS Code and capture the printed S/G trace, gain tables, final tree and test-case results. Replace the student details on the cover page before submission.